EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 30 - 305 - 1
	01/11/2022
MECHANICAL HARDWORK FOOT WEAR "SHOES"	

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SPECIFICATION
FOR
MECHANICAL HARDWORK
FOOT WEAR "SHOES"

Issue: NOV-2022/ Rev- 1

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MECHANICAL HARDWORK FOOT WEAR "SHOES"

1- Scope:

This specification defines the minimum technical requirements for Protect the feet from risk of cuts and fractures from heavy mechanical work (workshops – stores).

2- Applicable Codes and Standards:

The footwear should comply with the flowing standards:

- OHSA: 1910.136
- ISO: EN; 20345:
- ISO: EN: 20344;
- ASTM F2413 -O55
- EN 50321

Any provision of this specification differs from those of the standards listed above; the provision of this specification shall apply.

3- Definitions:

The mechanical Hard work footwear defined as:

Factory of natural flexible skin, 2 mm thick
The front of the shoe is covered with steel to protect the fingers
The sole contains a steel chip for penetration protection
The sole should be polyurethane

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4- General:

Thermal insulation
Impermeability (for water – for liquids)
With internal liner (sweat absorption – resistant to bacteria & fungi)
The sole of the shoe is flexible and non-skid
The sole of the shoe is cooked by injection

According to EN ISO 20345:2011:

- Any safety footwear shall be classified in accordance with Table 1.
- The safety footwear shall comply the basic requirements of EN ISO 20345: 2011, Table 2.
- The safety footwear may also have some additional features or requirements such as those listed in Table 3.

Table -1
Classification of footwear according to EN ISO 20345:2011.

Classification	Description
Class I	Footwear made from leather and other material, excluding all-rubber or all-polymeric footwear
Class II	All-rubber (i.e. entirely vulcanized) or all polymeric (i.e. entirely molded) footwear.

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Table-2
Some chosen basic requirements for safety footwear (for class II only) according to EN ISO 20345:2011.

Requirements		Subclause in 20345:2011	Class II
Design	Height of the upper	**5.2.2**	✓
	Seat region	**5.2.3**	✓
Whole footwear	Toe protection: General	**5.3.2.1**	✓
	Internal length of toecaps	**5.3.2.2**	✓
	Impact resistance	**5.3.2.3**	✓
	Compression resistance	**5.3.2.4**	✓
	Behavior of toecaps	**5.3.2.5**	✓
	Leakproofness	**5.3.3**	✓
	Specific ergonomic Features	**5.3.4**	✓
	Slip resistance	**5.3.5**	✓
Upper	Thickness	**5.4.2**	✓
	Tensile properties	**5.4.4**	✓
	Flexing resistance	**5.4.5**	✓
	Hydrolysis	**5.4.8**	✓
Outsole	Design	**5.8.1**	✓
	Tear strength	**5.8.2**	✓
	Abrasion resistance	**5.8.3**	✓

	Flexing resistance	**5.8.4**	✓
	Hydrolysis	**5.8.5**	✓

Note:

- **Resistance to hot contact (400 c^{o}) according to EN 20344 clause 8.7.1.2**

1. **Safety toecap**
   Built-in footwear component the wearer from impacts of an energy level of at least 200 Jand compression at a load of at least 15 KN.
2. **Water vapor permeability and coefficient**
   When tested in accordance with 6.6 and 6.8 of ISO 20344:2011, the water vapor permeability shall be not less than 0.8 mg /(cm^{2}·h) and the water vapor coefficient shall than 15 mg /cm^{2}.
3. **Abrasion resistance**
   When tested in accordance with ISO 20344:2011, 6.12, the lining shall not develop any holes before the following number of cycles has been performed.
4. **Water tested in accordance**
   When tested in accordance with 6.6 and 6.8 of ISO 20344:2011, the water vapor permeability shall be not less than 2,0 mg /(cm^{2}·h) and the water vapor coefficient shall be not less than 20 mg /cm^{2}.
5. **Water absorption and desorption**
   When tested in accordance with ISO 20344:2011,7.2 , the water absorption shall be not less than 70 mg / cm^{2} and the water desorption shall be not less than 80% of the water absorbed .
6. **Flexing resistance**
   When outsoles are tested in accordance with ISO 20344:2011,8.4, the cut growth shall be not greater than 4 mm before 30000 flex cycles. Spontaneous cracks are acceptable in the following circumstances.

a) Only the center of the tread area shall be assessed for cracking, i.e. cracks under the toecap zone shall be ignored.
b) Superficial cracks up to 0.5 mm deep shall be ignored.
c) Soles shall be deemed to be satisfactory if cracks are no deeper than 1.5 mm, no longer than **4** mm and no more than five in number.

7. Antistatic footwear

When measured in accordance with ISO 20344:2011, 5.10 after conditioning:

- In a dry atmosphere, the electrical resistance shall be above 100 **KΩ** and less than or equal to 1000 MΩ.
- In a wet atmosphere, the electrical resistance shall be above 100 **KΩ** and less than or equal to 1000 MΩ.

Note See ISO 20344:2011, 5.10.3.3 for determination of dry and wet atmospheres.

8. Marking

Each item of safety footwear shall be clearly and permanently marked, e.g. by embossing or branding, with the following:

a) Size;
b) Manufacturer's identification mark;
c) Manufacturer's type designation;
d) Year and at least quarter of manufacture;
e) Reference to the International Standard, i.e. ISO 20345:2011;
f) Symbol(s) from Table 2 and Table 18 appropriate to protection provided and/or, where applicable, the appropriate category (SB, S1 to S5), as described in Table 20 and 21.

Natural leather according to EN- ISO 203456

1. Fly toecap
2. Conductive foot wear.
3. Antistatic footwear

4. Water vapor permeability
5. Abrasion resistance
6. Water absorption and desorption
7. Flexing resistance
8. Water resistance.

Packaging:

The manufacturer shall define the packaging type suitable for transportation of the mechanical Hard work footwear.

Each pair of electrical insulating footwear shall pack in an individual container or package.

The outside of packaging shall be marked with the name of the manufacturer or supplier, classification, size and design.

Information to be supplied by the manufacturer